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SOFTWARE ENGINEERING

M01: Introduction to Software Engineering & Project Vision

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10 The floor is yours: QuickBite

Submission	Required coverage	Points
Checkpoint 1	M01–M02	20
Checkpoint 2	M01–M04	20
Checkpoint 3	M01–M06	20
Checkpoint 4	M01–M08	20
Checkpoint 5 — final project	M01–M10	20
Total	**10 modules**	**100**

MOODLE CHECKPOINTS**

Checkpoint 1
M01–M02

Checkpoint 2
M01–M04

Checkpoint 3
M01–M06

Checkpoint 4
M01–M08

Checkpoint 5
final project*

*final project submission - 1 week after our last class

*Always export your project before leaving the lab. A Moodle checkpoint is not the same thing as a progress backup.

Syllabus topic	Coverage
Project vision, lifecycle and feasibility	M01
Business, functional and non-functional requirements	M02
System functions and information flows	M03–M04
Reliability, availability, security and fault tolerance	M05
Physical infrastructure and deployment	M06
UML diagrams and design patterns	M03, M07–M08
CASE tools	Website modelling tools used in required diagram tasks
Project management and professional practice	M09
Testing methods and practical exercises	M10: test plan, six test cases and two manual walkthroughs

QuickBite project - criteria / requirements

1. **Work in teams of 3**

Submit one agreed project ZIP per team. Everyone should contribute and be able to explain the team's decisions.

2. **Each module is worth 10 points**

Each module has 5 criteria worth 0–2 points each, published with its instructions:

- 2 points: complete, clear and justified.
- 1 point: partly correct, but incomplete or unclear.
- 0 points: missing or fundamentally incorrect.

3. **Different answers can earn full marks**

We assess your reasoning, use of the project brief and consistency between your answers. There is no single ideal solution.

4. **Focus on required tasks**

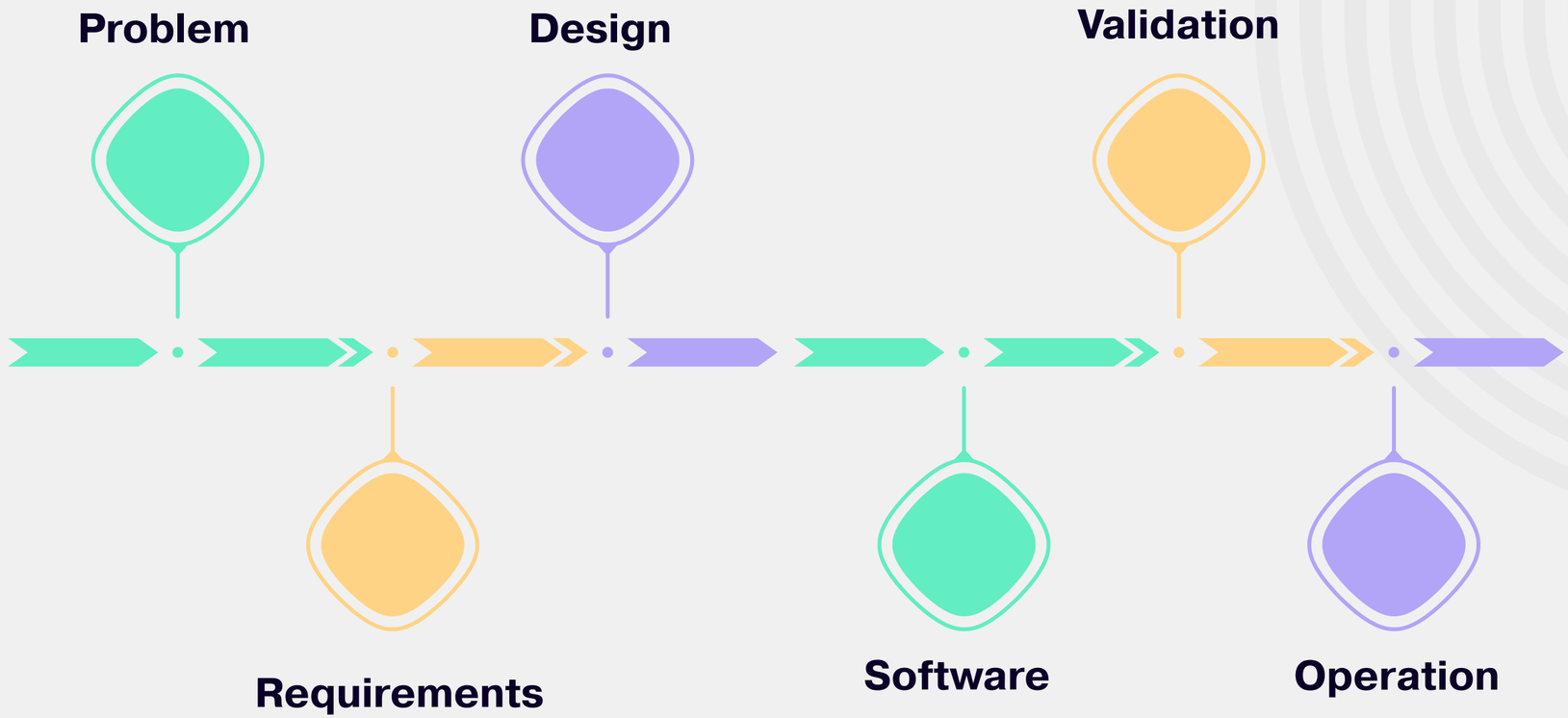
Optional tasks are extra practice. Skipping them causes no penalty; completing them adds no points.

5. **Submit your work through Moodle**

Save your answers on the website, export the latest project ZIP and upload it by the assignment deadline. "Saved" on the website confirms storage - it is not a grade.

6. **Project total: 100 points**

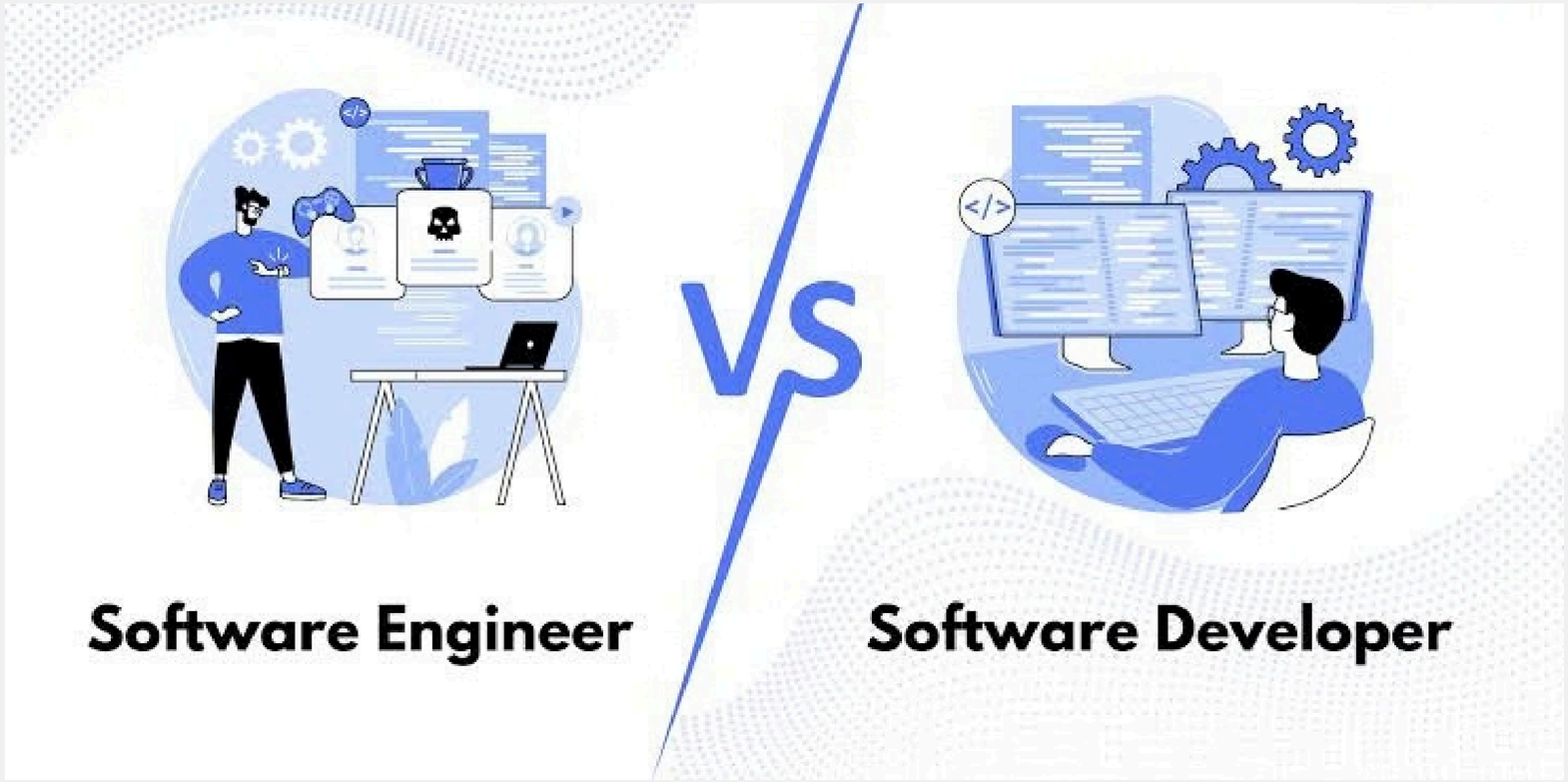
M01 is worth 10 points across Sessions 1 and 2 together. The project contributes to the laboratory component;



IDENTIFYING THE GAP

SOFTWARE ENGINEERING VS. SOFTWARE DEVELOPMENT

What’s the difference?



Software Engineering vs. Software Development

Focus on the complete system

Includes defining the problem

Requirements + design + implementation + testing

Usually team-based

Long-term maintainability

Many engineering artifacts

Focus on code

Usually solves a defined problem

Implementation-oriented

May be individual

Short-term correctness

Code is the main artifact

A program that works today is not necessarily a well-engineered software system.

Example: Car Rental Company

Scenario

A car rental company currently handles reservations through phone calls, email and spreadsheets.

- customers do not know which cars are available
- employees manually verify reservations
- double booking can happen
- cancellations are difficult to track
- managers lack reliable utilization information

Is the solution simply “build a booking website”?

Software Engineering Starts with the Problem

Request:
“We need an online calendar.”

Underlying problem:
“Customers and staff do not have a single reliable view of vehicle availability.”



Business problem

Customers cannot see which cars are available.
Branch employees confirm reservations manually using phone calls, emails and spreadsheets.

Stakeholder needs

Customer: “I want to know immediately whether a car is available.”
Branch Employee: “I need one reliable view of reservations and vehicle availability.”
Branch Manager: “I want fewer booking conflicts and less manual work.”
Fleet Manager: “I need vehicles under maintenance to be unavailable for booking.”

Requirements

The system shall allow customers to search for available vehicles for a selected location and rental period.
The system shall prevent confirmed reservations for the same vehicle during overlapping rental periods.
The system shall allow authorized staff to mark a vehicle as unavailable.

System design

Possible design decisions:
central availability service | reservation service | vehicle/fleet service | integration with payment provider | shared reservation database

Implementation

Examples:
availability search | reservation creation | cancellation handling | staff availability management | payment integration

Validation

We check whether the solution actually solves the original problem.
Examples:
- Can two customers reserve the same car for overlapping dates?
- Does a vehicle marked “maintenance” disappear from availability?
- Can staff see the same reservation data as the customer?
- Has average reservation handling time decreased?

Software Engineering Starts With the Problem*

***Notice that build a booking website never appears as the business problem.**

Business problem

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Software Engineering Starts With the Problem*

***Notice that build a booking website never appears as the business problem.**

What is a Software Product?



- programs
- configuration
- documentation
- data structures
- operational procedures
- interfaces
- test assets
- deployment information
- maintenance information

Quality of Professional Software

Maintainability
Can the system be changed safely?

Dependability & Security
Can users trust the system and its data?

Efficiency
Does it use time and resources appropriately?

Acceptability
Can intended users understand and use it?

Quality of Professional Software - Car Rental example

Maintainability
Can the system be changed safely?

→ add electric vehicles next year

Dependability & Security
Can users trust the system and its data?

→ protect customer and payment information

Efficiency
Does it use time and resources appropriately?

→ availability search should respond quickly

Acceptability
Can intended users understand and use it?

→ branch employees must be able to use it without lengthy training

Four Fundamental Software Activities

SPECIFICATION

What should the system do?

DESIGN & IMPLEMENTATION

How will we build it?

VALIDATION

Did we build **the right** system?

EVOLUTION

How will it **change** after release?

They exist in almost every software project, although their order and intensity differ.

Four Fundamental Software Activities - Car Rental example

SPECIFICATION

What should the system do?

→ Search available cars, prevent double booking, support reservations.

DESIGN & IMPLEMENTATION

How will we build it?

→ Design vehicle/reservation data and implement booking, payment and staff functions.

VALIDATION

Did we build **the right** system?

→ Test overlapping bookings, unavailable cars, failed payments and cancellations.

EVOLUTION

How will it **change** after release?

→ Add new branches, EV features, new payment providers or updated cancellation rules.

These activities are not necessarily four completely separate phases. In an iterative project, we may revisit specification, design and validation many times.

From Idea to Project Vision

A project vision should explain:

- what problem exists,
- who experiences it,
- what change the project aims to create,
- what success would look like.

Bad:

Build a React application with PostgreSQL.

Better:

Provide customers and branch employees with a reliable way to view vehicle availability and manage reservations without manual spreadsheet coordination.

Avoid too much technical detail.

A Good Problem Statement

[WHO] EXPERIENCES [PROBLEM], WHICH CAUSES [IMPACT].
THE PROJECT SHOULD IMPROVE [MEASURABLE OUTCOME] WITHOUT [IMPORTANT CONSTRAINT].

CAR RENTAL EXAMPLE

Customers and branch employees do not have a single reliable source of vehicle availability, causing booking conflicts and manual verification work. The proposed system should reduce booking errors and reservation handling time while supporting existing branch operations.

A problem statement should not contain the solution.

Stakeholders

A stakeholder is a person, group or external organization that can affect the system or is affected by it.

CUSTOMER

BRANCH EMPLOYEE

BRANCH MANAGER

FLEET MANAGER

FINANCE

PAYMENT PROVIDER

CUSTOMER SUPPORT

And many more…

A stakeholder is not necessarily a direct user.

Stakeholders also include people and organisations whose interests we must consider - even when they never open the application.

Stakeholder Needs Can Conflict

Customer: wants maximum flexibility to cancel.

Business: wants to avoid cars remaining unused after last-minute cancellations.

Which requirement should win?

- Customer → Fast booking
- Branch employee → Accurate availability
- Manager → High vehicle utilization
- Finance → Correct payments and deposits
- Fleet manager → Cars unavailable during maintenance
- Support → Clear booking history

Stakeholder Needs Can Conflict

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Engineering does not automatically “pick a side”. The conflict must be identified, negotiated and documented.

Stakeholders needs, concerns and questions - example

Need = desired outcome

I need reliable vehicle availability

Concern = possible difficulty / risk

I worry that outdated availability will cause double booking

Open question = information to clarify

How quickly must availability updates appear?

Define: In Scope vs. Out of Scope

In scope

Capabilities the project intends to deliver.

Out of scope

Capabilities explicitly not included.

Define: In Scope vs. Out of Scope - Car Rental example

In scope

Capabilities the project intends to deliver.

- **browse available vehicles,**
- **create reservation,**
- **cancel reservation,**
- **manage vehicle availability.**

Out of scope

Capabilities explicitly not included.

- **automatic dynamic pricing,**
- **vehicle GPS tracking,**
- **insurance claim processing.**

Facts, Assumptions, Constraints, Questions

Fact - explicitly supplied information

The company operates 8 branches

Assumption - unverified working proposition

Most customers have smartphones

Constraint - imposed limit on choices

Must use the existing payment provider

Open question - missing information to clarify

How long may a customer hold a reservation?

Assumptions and open questions should be made visible - hidden uncertainty becomes project risk.

Facts, Assumptions, Constraints, Questions - Car Rental example

Fact - explicitly supplied information

The company operates 8 branches

Reservations are currently handled manually.

Assumption - unverified working proposition

Most customers have smartphones

Customers will use online self-service.

Constraint - imposed limit on choices

Must use the existing payment provider

Existing payment provider must remain.

Open question - missing information to clarify

How long may a customer hold a reservation?

How long may a reservation be held before payment?

Assumptions and open questions should be made visible - hidden uncertainty becomes project risk.

BUSINESS OBJECTIVES

Objectives should describe outcomes, not features.

FEATURE:

IMPLEMENT ONLINE RESERVATION.

OBJECTIVE:

REDUCE AVERAGE RESERVATION HANDLING TIME FROM 8 MINUTES TO BELOW 3 MINUTES.

Baseline → Target → Measurement → Time period

Business Objectives - example

Objectives should describe outcomes, not features.

FEATURE:

IMPLEMENT ONLINE RESERVATION.

OBJECTIVE:

REDUCE AVERAGE RESERVATION HANDLING TIME FROM 8 MINUTES TO BELOW 3 MINUTES.

Objective should consist of:

- **an outcome,**
- **a metric and**
- **a target / direction**

Example: Reduce average staff reservation-handling time from an assumed 8 minutes to below 3 minutes within three months after rollout.

Outcome: less manual work.

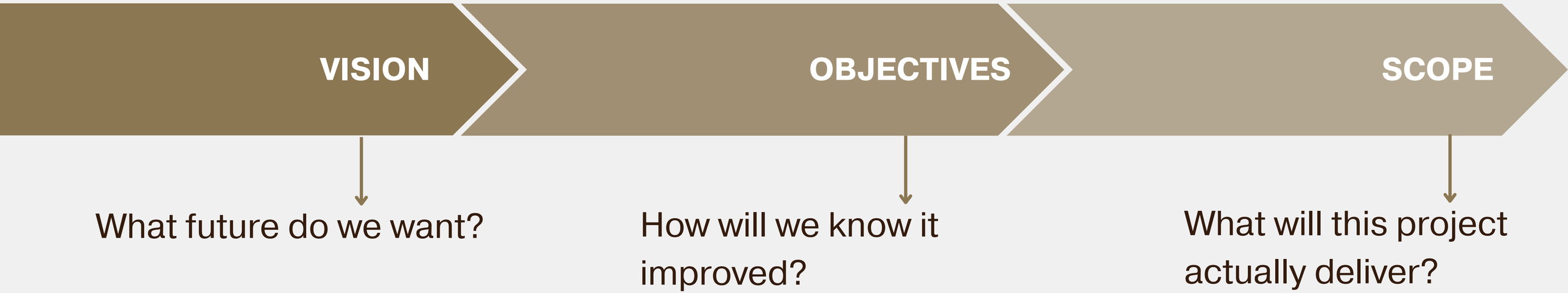
Metric: average staff minutes per handled reservation.

Measurement: record handling times for comparable reservation samples before/after rollout.

Objective: Outcome → Metric → Target / Direction

Vision → Objectives → Scope

PROBLEM: Why change?



Student Project: QuickBite

A fictional food-ordering and delivery platform connecting customers, restaurants and couriers

The website contains the full company brief, task instructions and required evidence.

M01 - Student Task - Teams of 3

Website module M01 - Tasks 1-5

Start / resume on the website

1. Open the QuickBite link in Moodle (or the address below).
2. Set Team ID and Current session to 1.
3. Read Project Brief, then M01 guidance and Tasks 1-5.
4. Write in M01 Report; use the stated save controls.

No starter ZIP, local server or Python installation is needed.

https://quickbite-workbench.vercel.app/

QuickBite: how to work, save and submit

***Work in a team of 3**. Use one agreed Team ID and produce one shared M01 project package. Discuss decisions together: each member should be able to explain the submitted work. The website contains the QuickBite company brief and task instructions; lecture slides explain theory using another company.*

1. Read the project information
Open Project Brief for the company background and project rules. Open M01 → Concepts / Student task for explanations, instructions and completion checklists.

2. Work directly on the website
Your workspace is in this browser. Write on task cards and click Save answer.
For Tasks 3–4, use Your project → Save stakeholders / Save scope.
Saved work is not shared automatically with teammates or the teacher.

3. Export before leaving
Open Save & Export → Save identity & download project ZIP.
Keep the ZIP outside the classroom computer.
Next class, open the same website and use
Save & Export → Import project with your latest ZIP.

M01:
4. Complete the required tasks
Session 1: Tasks 1–5. Session 2: Tasks 6 and 8.
Task 7 is optional: extra practice, no extra points and no penalty for skipping it.

5. M01 is worth 10 points in total
Five criteria, 0–2 points each.

M01 assessment 10 points total

Criterion	Points
Problem vision and explicit uncertainty	0–2
Stakeholders and conflict	0–2
Scope and two measurable objectives	0–2
Justified lifecycle choice	0–2
Feasibility three risks and decision	0–2

0 = absent or fundamentally incorrect;
1 = partly correct but incomplete or unclear;
2 = meets the checklist with clear, consistent reasoning. M01 is assessed as one module across both sessions. Optional exercises do not increase its maximum score. “Saved” confirms storage/structure, not correctness or a grade.

6. Submit one complete project ZIP on Moodle after each Checkpoint.
Do not edit the exported files manually or submit only M01.md. The ZIP contains your report (workspace/reports/M01.md), task answers and other data. Make corrections on the website, then export again.*
***If you want to update .md report file manually, copy it outside of the exported folder, then edit and upload additionally to Moddle’s Module folder if needed.**

END OF CLASS

One website. One team project.

Keep your ZIP.

https://quickbite-workbench.vercel.app

Save your evidence

Reports/code save in the browser. Save models and simulator runs using their save buttons.

Download your project ZIP

Save & Export > Save identity & download project ZIP.
Check Module deliverables.

Keep and share the agreed file

Store it outside the university computer. Submit on Moodle when requested; each backup is not automatically a submission.

Next class: return and import

Open the same website; import the latest ZIP if needed. Select the new Current session. Do not extract and start a server.

THANK YOU